

BEE 208 — Question 4: Fuse Selection Menu

Control Statement: switch | Faculty of Engineering, ATU

1. ALGORITHM

Step 1: Start

Step 2: Declare variable choice (int)

Step 3: Display the fuse selection menu (options 1 to 5)

Step 4: Prompt user to enter choice

Step 5: Read choice

Step 6: Use switch on choice

Step 7: Case 1 — display application for 5 A fuse

Step 8: Case 2 — display application for 10 A fuse

Step 9: Case 3 — display application for 13 A fuse

Step 10: Case 4 — display application for 20 A fuse

Step 11: Case 5 — display application for 32 A fuse

Step 12: Default — display 'Invalid choice'

Step 13: End

2. PSEUDOCODE

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BEGIN

DECLARE choice AS INTEGER

OUTPUT menu (1. 5A, 2. 10A, 3. 13A, 4. 20A, 5. 32A)

INPUT choice

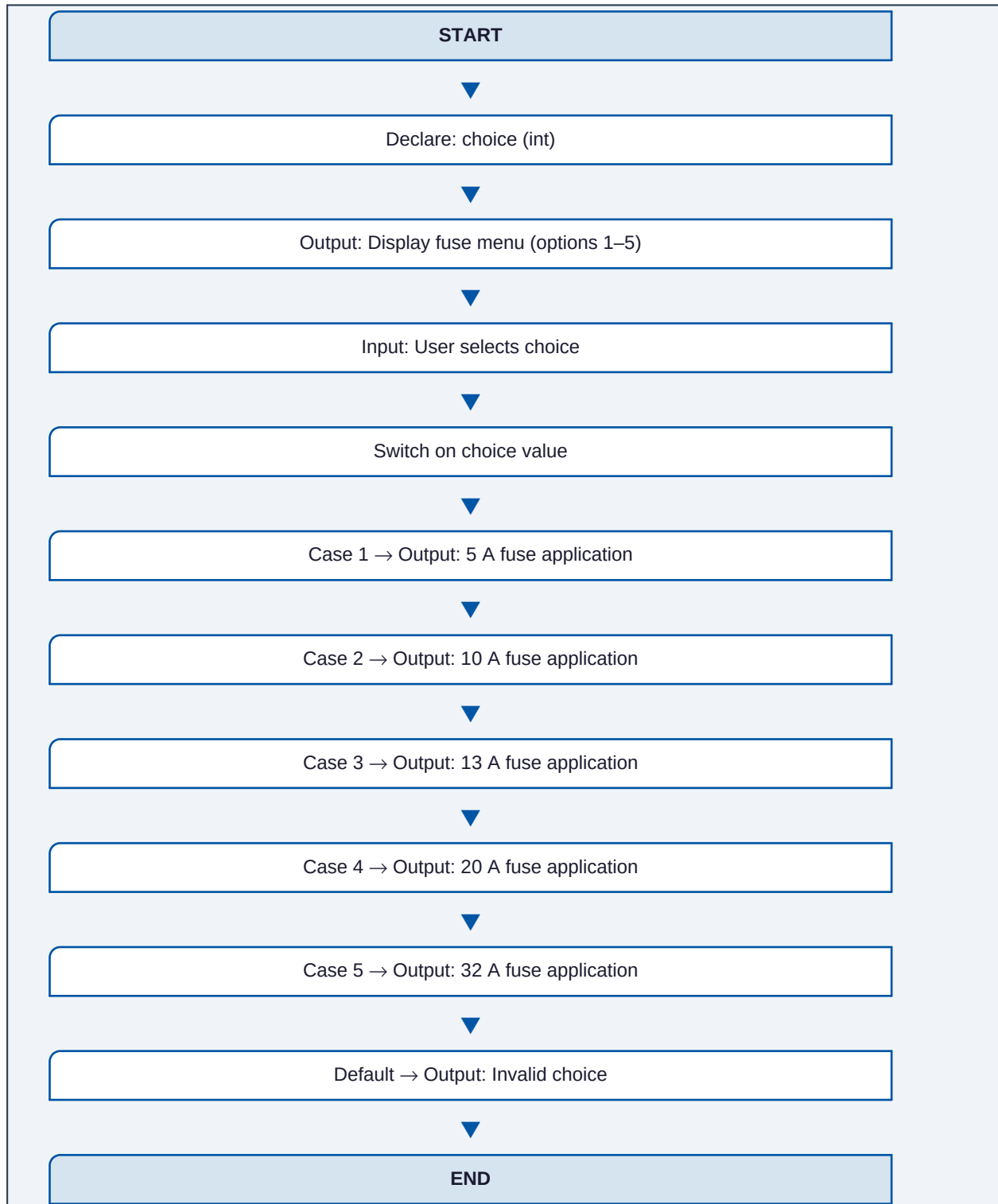
SWITCH choice

CASE 1: OUTPUT '5 A - Low-power circuits and lighting' BREAK
CASE 2: OUTPUT '10 A - General lighting circuits' BREAK
CASE 3: OUTPUT '13 A - Standard 13 A plug sockets' BREAK
CASE 4: OUTPUT '20 A - Immersion heaters' BREAK
CASE 5: OUTPUT '32 A - Cookers and electric showers' BREAK
DEFAULT: OUTPUT 'Invalid choice'

END SWITCH

END
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3. FLOWCHART (Structured Diagram)



4. TEST RESULTS

Test Item	Value / Description	Test Item	Value / Description
Input	choice = 1	Expected Output	5 A - Low-power circuits and lighting
Input	choice = 5	Expected Output	32 A - Cookers and electric showers
Input	choice = 9	Expected Output	Invalid choice

